

Azure Services Learning Resources

Quick Start Links

- [Azure Free Account Signup](#)
 - [Azure Portal](#)
 - [Microsoft Learn](#)
 - [Azure Documentation](#)
 - [Azure Samples GitHub](#)
 - [Azure CLI Docs](#)
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The 7 Buckets Overview

1. Foundation & Security
 2. Compute & Containers
 3. Storage & Databases
 4. Networking & Content Delivery
 5. DevOps & Automation
 6. AI & Machine Learning
 7. Monitoring & Cost
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BUCKET 1: Foundation & Security

Learning Links

- Azure Resource Hierarchy Overview:
<https://notes.kodekloud.com/docs/Microsoft-Azure-Security-Technologies-AZ-500/Enterprise-Governance/Understanding-Azure-hierarchy/page>
- Microsoft Entra ID (Azure AD) basics:
<https://learn.microsoft.com/en-us/entra/fundamentals/>
- Azure Role Based Access Control (RBAC):
<https://learn.microsoft.com/en-us/azure/role-based-access-control/overview>

- Azure Key Vault documentation:
<https://learn.microsoft.com/en-us/azure/key-vault/general/overview>
 - Azure Tags and Naming conventions:
<https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/azure-best-practices/resource-naming>
<https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/resource-name-rules>
<https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/azure-best-practices/resource-abbreviations>
 - Azure Budgets and spending alerts:
<https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/cost-management-alerts-monitor-usage-spending>
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Day One Setup Checklist

- ☐ Create Azure account
Setup guide: [📺 How to create FREE Azure Account](#)
 - ☐ Create first subscription
Tutorial: [📺 How to create an Azure Subscription](#)
 - ☐ Create a resource group
Tutorial: [📺 6. How to Create and Manage a Resource Group in Azure Portal | Step...](#)
 - ☐ Set spending alert (\$50 recommended)
Tutorial: [📺 AZ-900 Episode 37 | Azure Cost Management](#)
 - ☐ Understand Azure hierarchy
Management Groups → Subscriptions → Resource Groups → Resources
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Hands-on Tasks

- ☐ Create a resource group and deploy a sample resource
 - ☐ Create a service principal and assign a limited RBAC role
 - ☐ Create a Key Vault and store a secret
 - ☐ Retrieve a secret from Key Vault using CLI or SDK
 - ☐ Add tags to a resource group and view them in cost analysis
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Resources

Microsoft Learn Azure Fundamentals Learning Path:

<https://learn.microsoft.com/en-us/credentials/certifications/azure-fundamentals/>

Azure RBAC Labs:

https://microsoftlearning.github.io/AZ500-AzureSecurityTechnologies/Instructions/Labs/LAB_01_RBAC.html

Azure Security and Identity Fundamentals Labs:

<https://microsoftlearning.github.io/SC-900-Microsoft-Security-Compliance-and-Identity-Fundamentals/>

BUCKET 2: Compute & Containers

Learning Links

- Azure Virtual Machines fundamentals:
<https://learn.microsoft.com/en-us/azure/virtual-machines/overview>
 - Azure Functions overview:
<https://learn.microsoft.com/en-us/azure/azure-functions/functions-overview>
 - Azure Kubernetes Service (AKS) basics:
<https://learn.microsoft.com/en-us/azure/aks/what-is-aks>
 - Azure Container Apps documentation:
<https://learn.microsoft.com/en-us/azure/container-apps/>
 - Azure Container Instances basics:
<https://learn.microsoft.com/en-us/azure/container-instances/>
 - Azure App Service overview:
<https://learn.microsoft.com/en-us/azure/app-service/overview>
 - Azure Batch overview:
<https://learn.microsoft.com/en-us/azure/batch/>
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Core Services Notes

Azure Virtual Machines

Azure's IaaS compute service for running full operating systems and applications.
Used for traditional applications, databases, GPU workloads, and custom environments.

Azure Functions

Serverless compute platform triggered by HTTP, timers, storage events, or messaging systems.

You only pay for execution time.

Azure Kubernetes Service (AKS)

Managed Kubernetes platform.

Azure manages the control plane while you manage worker nodes.

Azure Container Apps

Serverless container hosting platform built on Kubernetes but fully managed.

Supports autoscaling and scale-to-zero.

Azure Container Instances

Fastest way to run a single container in Azure.

No orchestration, no scaling logic.

Azure App Service

Fully managed hosting for web apps and APIs.

Supports .NET, Python, Node.js, Java, PHP.

Azure Batch

Service for running large-scale parallel workloads.

Used for simulations, rendering, scientific workloads, and distributed tasks.

Hands-on Tasks

- ☐ Launch a Linux VM and connect via SSH
 - ☐ Deploy a Function triggered by HTTP request
 - ☐ Build Docker image and deploy container to Container Apps
 - ☐ Create AKS cluster and deploy a containerized application
 - ☐ Run a batch job using Azure Batch
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Resources

Azure Compute Learning Paths:

<https://learn.microsoft.com/en-us/training/modules/describe-azure-compute-networking-services/>

AKS Hands-on Labs:

https://github.com/mtrainings/Azure_Kubernetes_Service

Serverless Development Labs:

https://olohmann.github.io/azure-hands-on-labs/labs/01_serverless/serverless.html

BUCKET 3: Storage & Databases

Learning Links

- Azure Blob Storage overview:
<https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction>
- Azure Managed Disks documentation:
<https://learn.microsoft.com/en-us/azure/virtual-machines/managed-disks-overview>
- Azure Files overview:
<https://learn.microsoft.com/en-us/azure/storage/files/storage-files-introduction>
- Azure SQL Database documentation:
<https://learn.microsoft.com/en-us/azure/azure-sql/database/?view=azuresql>
- Azure Cosmos DB overview:
<https://azure.microsoft.com/en-us/blog/a-technical-overview-of-azure-cosmos-db/>
<https://learn.microsoft.com/en-us/azure/cosmos-db/overview>
- Azure PostgreSQL Flexible Server docs:
<https://learn.microsoft.com/en-us/azure/postgresql/>
- Azure Managed Redis documentation:
<https://learn.microsoft.com/en-us/azure/redis/>
- Azure AI Search overview:
<https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search?ta>

[bs=indexing%2Cquickstarts](#)

- Azure Synapse Analytics documentation:
<https://learn.microsoft.com/en-us/azure/synapse-analytics/>
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Core Services Notes

Azure Blob Storage

Object storage for unstructured data such as images, videos, logs, and backups.

Storage tiers:

Hot

Cool

Cold

Archive

Azure Managed Disks

Block storage attached to Azure VMs.

Disk types include Standard HDD, Standard SSD, Premium SSD, and Ultra Disk.

Azure Files

Managed file shares accessible using SMB or NFS protocols.

Azure SQL Database

Fully managed SQL Server platform.

Handles patching, backups, and high availability.

Azure SQL Managed Instance

Managed SQL Server with higher compatibility for on-prem SQL workloads.

Azure Cosmos DB

Globally distributed NoSQL database supporting multiple APIs.

Azure Database for PostgreSQL / MySQL

Managed open-source relational databases.

Azure Managed Redis

In-memory caching platform for high-performance applications.

Azure AI Search

Managed search service with support for vector search and AI-powered indexing.

Azure Synapse Analytics

Enterprise data warehouse and analytics platform.

Hands-on Tasks

- ☐ Create Blob Storage account and upload files
 - ☐ Configure storage lifecycle rules
 - ☐ Attach managed disk to VM and take snapshot
 - ☐ Create Azure SQL Database and connect from application
 - ☐ Create Cosmos DB container and perform CRUD operations
 - ☐ Run a query in Synapse Analytics
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BUCKET 4: Networking & Content Delivery

Learning Links

- Azure Virtual Network fundamentals:
<https://learn.microsoft.com/en-us/azure/networking/fundamentals/>
- Azure Load Balancer documentation:
<https://learn.microsoft.com/en-us/azure/load-balancer/>
- Azure Application Gateway overview:
<https://learn.microsoft.com/en-us/azure/application-gateway/overview>
- Azure Front Door documentation:
<https://learn.microsoft.com/en-us/azure/frontdoor/>
- Azure Traffic Manager overview:
<https://learn.microsoft.com/en-us/azure/traffic-manager/traffic-manager-overview>

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- Azure Private Endpoint documentation:
<https://learn.microsoft.com/en-us/azure/private-link/private-endpoint-overview>
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Core Services Notes

Azure Virtual Network (VNet)

Private network environment in Azure.
Resources communicate securely within VNets.

Azure Load Balancer

Layer 4 load balancing for TCP and UDP traffic.

Azure Application Gateway

Layer 7 load balancing with HTTP awareness.
Includes Web Application Firewall.

Azure Front Door

Global edge network for routing traffic across regions.
Includes CDN and WAF capabilities.

Azure Traffic Manager

DNS-based global traffic routing.

Azure Private Endpoint

Allows private connectivity to Azure services within a VNet.

Hands-on Tasks

- ☐ Create VNet and subnets
- ☐ Configure firewall rules
- ☐ Deploy Application Gateway
- ☐ Enable Azure Front Door for global routing
- ☐ Create private endpoint for storage account

BUCKET 5: DevOps & Automation

Learning Links

- Azure Bicep documentation:
<https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/>
- Terraform Azure provider documentation:
<https://learn.microsoft.com/en-us/azure/developer/terraform/>
<https://developer.hashicorp.com/terraform/tutorials/azure-get-started>
- Azure DevOps Pipelines overview:
<https://learn.microsoft.com/en-us/azure/devops/pipelines/get-started/what-is-azure-pipelines?view=azure-devops>
- GitHub Actions for Azure:
<https://learn.microsoft.com/en-us/azure/developer/github/github-actions>
- Azure Service Bus documentation:
<https://learn.microsoft.com/en-us/azure/service-bus-messaging/>
- Azure Event Grid documentation:
<https://learn.microsoft.com/en-us/azure/event-grid/>
- Azure Event Hubs documentation:
<https://learn.microsoft.com/en-us/azure/event-hubs/>
- Azure Logic Apps overview:
<https://learn.microsoft.com/en-us/azure/logic-apps/logic-apps-overview>

Core Services Notes

Azure Bicep

Infrastructure as Code language designed specifically for Azure.

Terraform on Azure

Multi-cloud Infrastructure as Code tool widely used in enterprise environments.

Azure DevOps Pipelines

CI/CD pipelines integrated with Azure services.

GitHub Actions

CI/CD automation platform commonly used with Azure deployments.

Azure Service Bus

Enterprise-grade messaging service with guaranteed delivery.

Azure Event Grid

Event routing service for infrastructure events.

Azure Event Hubs

High-throughput data ingestion service.

Azure Logic Apps

Low-code automation platform connecting cloud and SaaS services.

Hands-on Tasks

- ☐ Deploy infrastructure using Bicep template
 - ☐ Provision infrastructure using Terraform
 - ☐ Create CI/CD pipeline with GitHub Actions
 - ☐ Publish and consume messages from Service Bus
 - ☐ Create event-driven workflow using Event Grid
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BUCKET 6: AI & Machine Learning

Learning Links

- Azure OpenAI documentation:
<https://learn.microsoft.com/en-us/azure/foundry/openai/reference>
- Azure Machine Learning overview:
<https://learn.microsoft.com/en-us/azure/machine-learning/?view=azureml-api-2>

- Azure AI Services documentation:
<https://docs.azure.cn/en-us/ai-services/>
 - Azure AI Foundry documentation:
<https://learn.microsoft.com/en-us/azure/foundry/>
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Core Services Notes

Azure OpenAI Service

Provides access to OpenAI models such as GPT, Whisper, and DALL-E through Azure infrastructure.

Azure Machine Learning

End-to-end platform for building, training, and deploying machine learning models.

Azure AI Services

Pre-built AI APIs for vision, speech, language, and document analysis.

Includes:

Computer Vision

Document Intelligence

Speech Services

Language Services

Translator

Azure AI Foundry

Unified platform for building production AI applications.

Supports RAG pipelines, agent systems, and AI orchestration workflows.

Hands-on Tasks

- ☐ Call Azure OpenAI API to build chatbot
 - ☐ Deploy ML model using Azure Machine Learning endpoint
 - ☐ Use Computer Vision API to analyze images
 - ☐ Extract data from documents using Document Intelligence
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BUCKET 7: Monitoring & Cost

Learning Links

- Azure Monitor documentation:
<https://learn.microsoft.com/en-us/azure/azure-monitor/>
 - Application Insights overview:
<https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview>
 - Microsoft Sentinel documentation:
<https://learn.microsoft.com/en-us/azure/sentinel/>
 - Microsoft Defender for Cloud overview:
<https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-cloud-introduction>
 - Azure Cost Management documentation:
<https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-cloud-introduction>
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Core Services Notes

Azure Monitor

Central observability service providing metrics, logs, dashboards, and alerts.

Application Insights

Application performance monitoring and distributed tracing.

Microsoft Sentinel

Cloud-native SIEM platform for threat detection and incident response.

Microsoft Defender for Cloud

Security posture management and vulnerability scanning.

Azure Cost Management

Cost tracking, budgets, and optimization recommendations.

Hands-on Tasks

- ☐ Create monitoring dashboard using Azure Monitor
 - ☐ Enable Application Insights on a web application
 - ☐ Configure budget alerts
 - ☐ Analyze cost usage with Cost Analysis
 - ☐ Enable Defender for Cloud recommendations
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The 3 Portfolio Projects

Project 1: Secure Static Website with CDN

Services Used:

Blob Storage

Azure Front Door

Web Application Firewall

Goal:

Host static website with global CDN and security.

Coding:

Minimal.

Project 2: Containerized API with AKS and CI/CD

Services Used:

Azure Kubernetes Service

Azure Container Registry

GitHub Actions

Goal:

Deploy containerized application with automated CI/CD.

Coding:

Dockerfile + Kubernetes manifests.

Project 3: Three-Tier E-Commerce Architecture

Services Used:

AKS

Azure SQL Database

Azure Managed Redis

Azure Front Door

Azure Storage

Goal:

Deploy scalable three-tier architecture.

Coding:

Microservices containers + Kubernetes manifests.

Cost Management (Link Slots)

Azure Free Tier reference:

Budget alerts guide:

<https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/cost-mgt-alerts-monitor-usage-spending>

Cost management documentation:

<https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/>

Cleanup / teardown guide:

<https://learn.microsoft.com/en-us/cli/azure/azure-cli-sp-tutorial-8?view=azure-cli-latest>

Communities

- [r/AZURE](#)
- [Microsoft Tech Community](#)
- [Azure Developer Community](#)